

Symbols

α_{s+}	The critical angle of attack at which stall occurs for positive angle of attack.
α_{s-}	The critical angle of attack at which stall occurs for negative angle of attack.
α	The angle of attack.
β	The sideslip angle.
$\delta\alpha$	The size of region where normal aerodynamics transition to post-stall aerodynamics.
$\delta\eta E_l$	The commanded tilt of the left engine.
$\delta\eta E_r$	The commanded tilt of the right engine.
δA	The deflection of the ailerons.
δE	The deflection of the elevator.
δN_l	The rev/s of the left propeller.
δN_r	The rev/s of the right propeller.
δR	The deflection of the rudder.
ηE_l	The tilt of the left engine.
ηE_r	The tilt of the right engine.
ηE	The symmetric tilt of both engines.
ω_{emax}	The maximum angular velocity of the servo+nacelle system.
$\bar{c}$	Mean aerodynamic chord.
ϕ	The roll of the craft on body axis x .
ψ	The yaw of the craft on body axis z .
ρ	The density of air that the craft is in.
τ_ϕ	The moment acting around the roll axis.
τ_ψ	The moment acting around the yaw axis.
τ_θ	The moment acting around the pitch axis.
τ_{Aero}	The moment acting on the craft due to aerodynamic forces.
τ_{Diff}	The moment acting on the craft due to differential rotation speed of propellers.
τ_{Thrust}	The moment acting on the craft due to thrust forces.
τ_e	The time constant representing the servo+nacelle response.
θ	The pitch of the craft on body axis y .
ξ	The state vector of the craft.
b	Wingspan.
$C_{D,max}$	The maximum value of the drag coefficient.
$C_{d,pre-stall}(\alpha)$	The pre-stall drag coefficient.
$C_{l,max}$	The maximum value of the lift coefficient.
$C_{l,pre-stall}(\alpha)$	The pre-stall lift coefficient.

C_{l_β}	The dihedral effect.
$C_{l_{\delta_A}}$	Aileron effectiveness.
$C_{l_{\delta_R}}$	Rudder-induced roll.
C_{l_r}	Yaw rate induced roll.
C_{l_p}	Roll damping.
C_{l_0}	Roll moment at trim.
C_{m_α}	Static longitudinal stability.
$C_{m_{\delta_E}}$	Elevator effectiveness.
C_{m_0}	Pitch moment at trim.
C_{m_q}	Pitch damping.
C_{n_β}	Weathercock stability.
$C_{n_{\delta_A}}$	Adverse yaw.
$C_{n_{\delta_R}}$	Rudder effectiveness.
C_{n_o}	Yaw moment at trim.
C_{n_p}	Roll-rate-induced yaw.
C_{n_r}	Yaw damping.
C_T	The thrust coefficient of the craft's propeller.
$C_d(\alpha)$	The coefficient of drag.
$C_l(\alpha)$	The coefficient of lift.
$C_y(\beta)$	The side-sweep coefficient.
D_{prop}	The diameter of the craft's propeller.
D	The force of drag acting on the craft along the x axis.
E_{Height}	The distance between the engines and CoM in the z axis..
E_{Span}	The distance between the engines and CoM in the y axis..
E_y	The distance between the CoM and the engine in y axis.
E_z	The distance between the CoM and the engine rotation axis in z axis.
F_{Aero}	The aerodynamical forces acting on the craft, includes F_{Aero_x} , F_{Aero_y} , F_{Aero_z} to divide into axes.
F_{Thrust}	The thrust forces acting on the craft, includes F_{Thrust_x} , F_{Thrust_y} , F_{Thrust_z} to divide into axes.
F_{Wind}	The aerodynamical forces acting on the craft, includes within the wind frame.
g_b	The gravitational forces acting on the craft.
K_q	The torque coefficient of the craft's propeller.
L	The force of lift acting on the craft.
m_{craft}	The mass of the craft.
p	The angular velocity of the craft along body axis x .
q_∞	Dynamic pressure.
q	The angular velocity of the craft along body axis y .
r	The angular velocity of the craft along body axis z .
S	Wing reference area.
S	The surface area of the wing.
u	The velocity in axis x of the craft within its body coordinates.
V_a	Airspeed.

v	The velocity in axis y of the craft within its body coordinates.
w	The velocity in axis z of the craft within its body coordinates.
x_E	The x position of the craft in the global coordinates.
X	The sum of all forces acting along axis x .
y_E	The y position of the craft in the global coordinates.
Y	The force of drag acting on the craft along the y axis.
Y	The sum of all forces acting along axis y .
z_E	The z position of the craft in the global coordinates.
Z	The sum of all forces acting along axis z .

Chapter 1

Dynamics

1.1 Overview

When considering a standard UAV dynamics the plant can be simplified to a 6 DoF rigid body where forces act on a point moving within space. In case of 2 engine tilt rotor design the system expands to a 6 + 2 DoF, where the two additional degrees of freedom represent the tilt of left and right engine nacelle. The entire system can still be simplified with following assumptions:

- The plant is a ideal rigid body, meaning the body maintains constant geometry. The angling of the nacelles and deflection of the control surfaces will be only considered in form of forces and torques acting on the body.
- The torque acting in the Y-axis generated by rotation of engine nacelles is negligible.
- The rotational complexity is simplified into a 3 x 3 mass moment of inertia tensor rotating around the center of mass.
- The mass of the plant remains constant.
- The body of the aircraft is symmetric in X - Z Plane, simplifying the inertia matrix.

The dynamics equations will be derived with Newton - Euler method as it allows an easy, step by step process to account for forces acting in the UAV. For a 6-DoF system the translational and rotational equation of motion within Newton - Euler method are as follow:

- Translational motion - the linear accelerations of the center of mass of the plant follow $F_b = m(\dot{V}_b + \omega_b \times V_b)$, where F_b are the forces acting on the body, V_b are the velocities within body frame and ω_b are the angular velocities. The linear accelerations are derived from the Newton's Law.
- Rotational motion - the rotation of the body that accounts for gyroscopic coupling through $M_b = I\dot{\omega}_b + \omega_b \times (I\omega_b)$, where M_b are the torques acting on the plant and I that is the 3 x 3 inertia matrix.

The dynamic of the plant will follow the NED coordinate convention with Z-axis pointing down and following the NEDs right handed rule and the GP-LV (Global Position - Local Velocity) convention where velocities are calculated within the body frame of the plant and positions exists within the global frame where point (0, 0, 0) is defined as position of the plant at $t = 0$.

The state-vector of the plant will be described as:

$$\xi = \begin{bmatrix} R_1^T & R_2^T & v_1^T & v_2^T & E_t^T \end{bmatrix}^T = \begin{bmatrix} x_E & y_E & z_E & \phi & \theta & \psi & u & v & w & p & q & r & \eta E_r & \eta E_l \end{bmatrix}^T \quad (1.1)$$

Where:

- $R_1^T = \begin{bmatrix} x_E & y_E & z_E \end{bmatrix}^T$ is the global position vector of the craft
- $R_2^T = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T$ is the vector containing the three euler angles (roll, pitch, yaw) in the inertial frame
- $v_1^T = \begin{bmatrix} u & v & w \end{bmatrix}^T$ is the velocity vector within the body frame of the craft
- $v_2^T = \begin{bmatrix} p & q & r \end{bmatrix}^T$ is the vector containing the angular velocities.
- $E_t^T = \begin{bmatrix} \eta E_r & \eta E_l \end{bmatrix}^T$ is the vector containing the throttle of the engines.

The orientation of each are showcased in 1.1, 1.2 and 1.3.

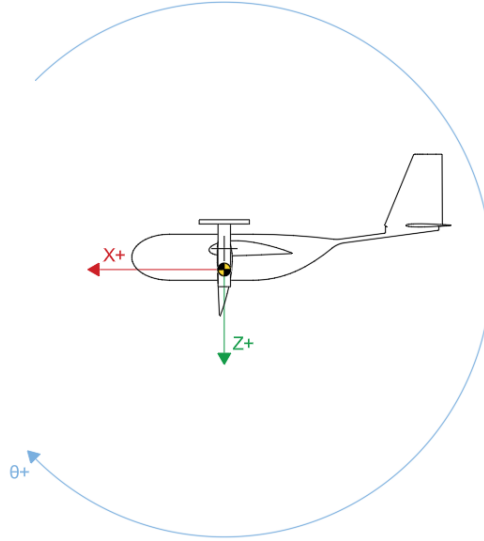


FIGURE 1.1: The axes within the body frame as visible from the side.

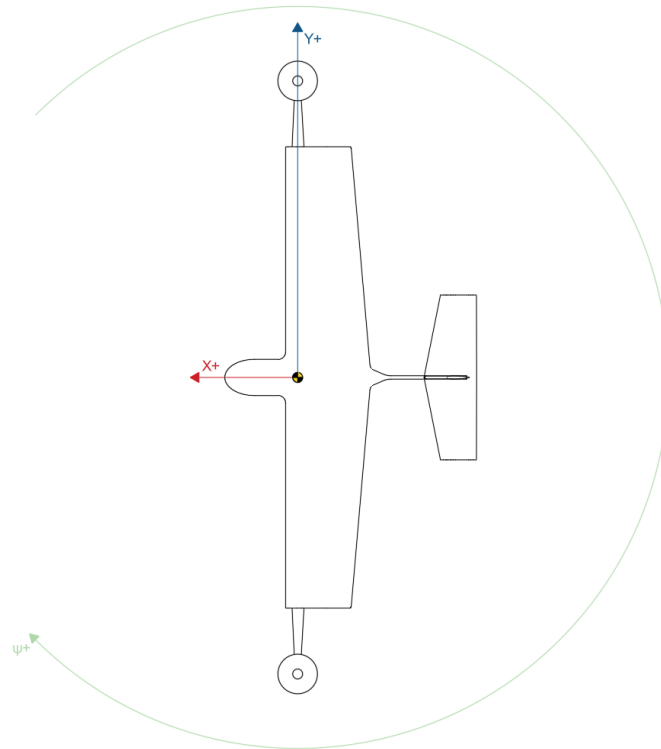


FIGURE 1.2: The axes within the body frame as visible from the top.

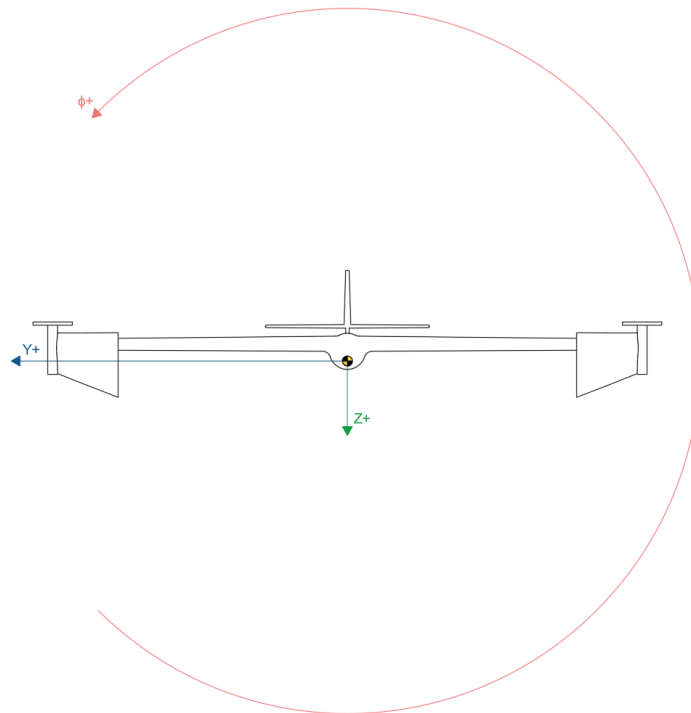


FIGURE 1.3: The axes within the body frame as visible from the front.

1.2 The delayed response of the engine tilting mechanism.

While the small size of the systems this model is intended to be used for means that response from the tilt mechanism is near instantaneous it also means that response of the airframe to any disturbance is also extremely fast and the fast actuator dynamics become unlikely to achieve the 10:1 time scale separation ratio in relation to slow plant dynamics where the actuators dynamics can be safely omitted. To model the delay of the tilting mechanism a rate-limited first-order model will be used in form:

$$\dot{\theta} = clip(\frac{1}{\tau}(\theta_{cmd} - \theta), -\omega_{max}, +\omega_{max}) \quad (1.2)$$

This results in two equations:

$$\eta \dot{E}_r = clip(\frac{1}{\tau_e}(\delta \eta E_r - \eta E_r), -\omega_{emax}, +\omega_{emax}) \quad (1.3)$$

$$\eta \dot{E}_l = clip(\frac{1}{\tau_e}(\delta \eta E_l - \eta E_l), -\omega_{emax}, +\omega_{emax}) \quad (1.4)$$

1.3 Forces acting on the model

When studying the movement of rigid bodies within the framework of the Newton - Euler method the accelerations of the object are obtained from the forces acting on the craft via Newtons 2nd law. But because the craft will be operating within a rotating, body-fixed frame the kinematic transport theorem needs to be used resulting in Newtons 2nd law being expressed as $\vec{F} = m(\frac{d\vec{v}}{dt}) + m(\vec{\omega} \times \vec{v})$. When solving this equation for $\frac{d\vec{v}}{dt}$ the forces translate to acceleration in following way:

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} rv - qw + \frac{X}{m_{craft}} \\ pw - ru + \frac{Y}{m_{craft}} \\ qu - pv + \frac{Z}{m_{craft}} \end{bmatrix} \quad (1.5)$$

The sum of forces acting on the craft consists of the aerodynamic forces, thrust forces and gravity:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} F_{Thrustx} + F_{Aerox} \\ F_{Thrusty} + F_{Aero y} \\ F_{Thrustz} + F_{Aero z} \end{bmatrix} + m_{craft} g_b \quad (1.6)$$

1.4 Force of gravity

The force of gravity within the inertial earth frame takes form $g_E = \begin{bmatrix} 0 & 0 & g \end{bmatrix}^T$. To apply it withing the body frame of the craft it needs to be multiplied by a rotation matrix resulting in form:

$$g_b = g \begin{bmatrix} -\sin\theta \\ \sin\phi\cos\theta \\ \cos\phi\cos\theta \end{bmatrix} \quad (1.7)$$

1.5 Force of thrust

The force of thrust generated by the propellers can be described in relation to the rotation speed of the propellers as:

$$T(N) = C_T \rho N^2 D_{prop}^4 \quad (1.8)$$

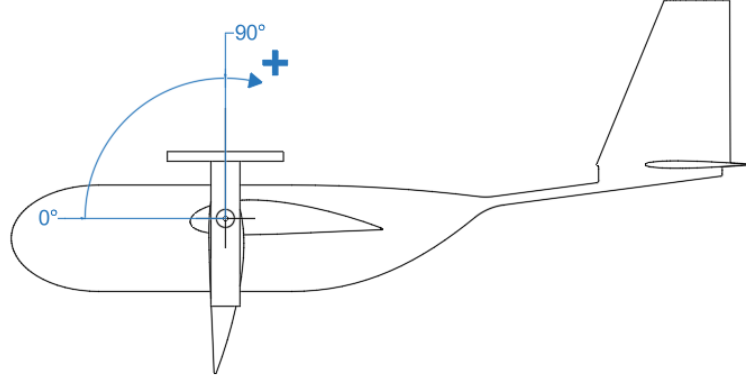


FIGURE 1.4: The angle of engine tilt.

The D_{prop} and C_T are parameters of the crafts construction. The air density ρ can also be treated as constant due to legal limits on drone flight altitude. During high speed flight the efficiency of propellers drops but due to limited speeds of drones this will also be simplified.

The forces acting on the craft via engines can be described as:

$$F_{Thrustx} = T(\delta N_r)\cos\eta E_r + T(\delta N_l)\cos\eta E_l \quad (1.9)$$

$$F_{Thrusty} = 0 \quad (1.10)$$

$$F_{Thrustz} = -T(\delta N_r)\sin\eta E_r - T(\delta N_l)\sin\eta E_l \quad (1.11)$$

The forces acting in each axis depend on tilting of each engine. As each engine needs to be able to tilt independently to allow for control over the craft the force of each needs to be calculated independently and summed. The force in axis z achieves maximum at 90° tilt and zero at 0° tilt. The force in axis x behaves inversly. At no engine position the force acts in y axis. As the z axis points downwards the forces acting on it must be inverted.

1.6 Aerodynamic Forces

Aerodynamic forces act withing the airflow frame that is rotated by α and β in relation to the body frame. Within this frame the forces acting upon the plant are:

- Lift L - Acts perpendicular to the relative airspeed in the symmetry plane, countering gravity.
- Drag D - Acts parallel and opposite to the relative airspeed, resisting forward motion.
- Sideforce Y - Acts along the lateral axis introducing sideslips and crosswinds.

The aerodynamic forces act in relation to the True Air Speed that can be defined as the magnitude of the relative velocity vector between the aircraft center of mass and the surrounding mass of air. This can be acquired from equation:

$$V_a = \sqrt{u^2 + v^2 + w^2} \quad (1.12)$$

The angle α is the angle between longitudinal body axis x and the relative wind velocity vector projected onto planes x - z symmetry plane. The angle α affects lift and longitudinal stability of the craft. It can be acquired from relation of the aircraft w and u velocities.

$$\alpha = \arctan2(w, u) \quad (1.13)$$

The angle β is defined as the angle between the total relative wind velocity vector and the aircraft's plane of symmetry (x - z plane) and can be acquired from:

$$\beta = \text{atan2}(v, \sqrt{w^2 + u^2}) \quad (1.14)$$

It must be noted that when airspeed becomes 0 at hover the α and β becomes physically meaningless and behavior of these equations is undefined. When implementing this model a convention for zero air speed should be implemented. For a VTOL craft accurate modeling of post stall behavior at extreme α angles becomes important and thus must be modeled. To model the aerodynamic forces in post stall conditions a modified Viterna-Corrigan model will be used, where C_l becomes $\frac{1}{2}C_{d,max}\sin(2\alpha)$ and C_d becomes $C_{D,max}\sin^2\alpha$. The models will be blended with standard, table derived C_l and C_d coefficients via function:

$$\sigma = \frac{1}{2}(2 + \tanh(\frac{\alpha - \alpha_{s+}}{\delta\alpha}) + \tanh(\frac{\alpha - \alpha_{s-}}{-\delta\alpha})) \quad (1.15)$$

The lift and drag coefficients thus become:

$$C_l(\alpha) = (1 - \sigma)(C_{l,pre-stall}(\alpha)) + \sigma(\frac{1}{2}C_{D,max}\sin(2\alpha)) \quad (1.16)$$

$$C_d(\alpha) = (1 - \sigma)(C_{d,pre-stall}(\alpha)) + \sigma(C_{D,max}\sin^2\alpha) \quad (1.17)$$

The aerodynamic within the wind frame can be defined as:

$$L = \frac{1}{2}\rho V_a^2 S C_l(\alpha) \quad (1.18)$$

$$D = \frac{1}{2}\rho V_a^2 S C_d(\alpha) \quad (1.19)$$

$$Y = \frac{1}{2}\rho V_a^2 S C_y(\beta) \quad (1.20)$$

These can be organized within a vector F_{Wind} . Due to force D acting opposing to the velocity u and force L acting opposing to the gravity these have to be inverted.

$$F_{Wind} = \begin{bmatrix} -D \\ Y \\ -L \end{bmatrix} \quad (1.21)$$

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To apply these forces to the body frame the forces have to be rotated by α and β :

$$F_{Aero} = R_y(\alpha)R_z(\beta)F_{Wind} \quad (1.22)$$

Where the transformation matrix is:

$$R_y(\alpha)R_z(\beta) = \begin{bmatrix} \cos\alpha\cos\beta & -\cos\alpha\sin\beta & -\sin\alpha \\ \sin\beta & \cos\beta & 0 \\ \sin\alpha\cos\beta & -\sin\alpha\sin\beta & \cos\alpha \end{bmatrix} \quad (1.23)$$

Applying these rotation results in component forces:

$$F_{Aerox} = -D\cos\alpha\cos\beta - Y\cos\alpha\sin\beta + L\sin\alpha \quad (1.24)$$

$$F_{Aero y} = -D\sin\beta + Y\cos\beta \quad (1.25)$$

$$F_{Aero z} = -D\sin\alpha\cos\beta - Y\sin\alpha\sin\beta - L\cos\alpha \quad (1.26)$$

1.7 Torques acting on the model

Within the Newton Euler method the torques are included via the Euler's Rotational Equation.

$$\tau = I\dot{\omega} + \omega \times (I\omega) \quad (1.27)$$

The two distinct parts of the equation account for the torque required to produce angular acceleration ($I\dot{\omega}$) and changing direction of the angular momentum vector due to rotation of the body frame in form of the cross product ($\omega \times (I\omega)$).

To use it within a model the vector cross product must be expanded with the inertial matrix of the aircraft and the resulting equation must be solved for $\dot{\omega}_x$, $\dot{\omega}_y$ and $\dot{\omega}_z$

$$I = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{zx} & 0 & I_{zz} \end{bmatrix} \quad (1.28)$$

This inertia matrix is the result of aircraft's symmetry in the x - z plane. This reduces products of inertia I_{xy} and I_{yz} to zero but leaves a dynamic coupling between roll and yaw through their accelerations. When this matrix expands the equation it becomes:

$$\tau_x = I_{xx}\dot{\omega}_x - I_{xz}\dot{\omega}_z + (I_{zz} - I_{yy})\omega_y\omega_z - I_{xz}\omega_x\omega_y \quad (1.29)$$

$$\tau_y = I_{yy}\dot{\omega}_y + (I_{xx} - I_{zz})\omega_x\omega_z + I_{xz}(\omega_x^2 - \omega_z^2) \quad (1.30)$$

$$\tau_z = -I_{xz}\dot{\omega}_x + I_{zz}\dot{\omega}_z + (I_{yy} - I_{xx})\omega_x\omega_y + I_{xz}\omega_y\omega_z \quad (1.31)$$

Computation of $\dot{\omega}_y$ is trivial due to simple coupling with other axes resulting in:

$$\dot{\omega}_y = \frac{\tau_y + (I_{zz} - I_{xx})\omega_x\omega_z - I_{xz}(\omega_x^2 - \omega_z^2)}{I_{yy}} \quad (1.32)$$

The equations for x and z axis share both $\dot{\omega}_x$ and $\dot{\omega}_z$ resulting in equations in form:

$$\dot{\omega}_x = \frac{I_{zz}L' + I_{xz}N'}{I_{xx}I_{zz} - I_{xz}^2} \quad (1.33)$$

$$\dot{\omega}_z = \frac{I_{xz}L' + I_{xx}N'}{I_{xx}I_{zz} - I_{xz}^2} \quad (1.34)$$

Where:

$$\begin{bmatrix} L' \\ N' \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xz} \\ -I_{xz} & I_{zz} \end{bmatrix} \begin{bmatrix} \dot{\omega}_x \\ \dot{\omega}_z \end{bmatrix} \quad (1.35)$$

Resulting in:

$$L' = \tau_\phi - (I_{zz} - I_{yy})qr + I_{xz}pq \quad (1.36)$$

$$N' = \tau_\psi - (I_{yy} - I_{xx})pq - I_{xz}qr \quad (1.37)$$

These equations can be ordered into a simplified form where most inertial terms are packed into coefficients:

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} c_1pq + c_2rq + c_7\tau_\phi + c_9\tau_\psi \\ c_3(p^2 - r^2) + c_4pr + c_8\tau_\theta \\ c_5pq + c_6rq + c_9\tau_\phi + c_{10}\tau_\psi \end{bmatrix} \quad (1.38)$$

Where:

$$\begin{aligned} c_1 &= \frac{-I_{zz}I_{xz} - I_{xz}(I_{xx} - I_{yy})}{I_{xz}^2 - I_{xx}I_{zz}} \\ c_2 &= \frac{-I_{zz}(I_{yy} - I_{zz}) + I_{xz}^2}{I_{xz}^2 - I_{xx}I_{zz}} \\ c_3 &= -\frac{I_{xz}}{I_{yy}} \\ c_4 &= -\frac{I_{xx} - I_{zz}}{I_{yy}} \\ c_5 &= \frac{-I_{xz}^2 - I_{xx}(I_{xx} - I_{yy})}{I_{xz}^2 - I_{xx}I_{zz}} \\ c_6 &= \frac{I_{xx}I_{xz} - I_{xz}(I_{yy} - I_{zz})}{I_{xz}^2 - I_{xx}I_{zz}} \\ c_7 &= -\frac{I_{zz}}{I_{xz}^2 - I_{xx}I_{zz}} \\ c_8 &= \frac{1}{I_{yy}} \\ c_9 &= -\frac{I_{xz}}{I_{xz}^2 - I_{xx}I_{zz}} \\ c_{10} &= -\frac{I_{xx}}{I_{xz}^2 - I_{xx}I_{zz}} \end{aligned}$$

The sum of torques acting on the craft consists of the aerodynamic torques, torques resulting from thrust and torques as result of differential angular velocities of the spinning propellers:

$$\begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} = \begin{bmatrix} \tau_{Aerox} + \tau_{Thrustx} + \tau_{Diffx} \\ \tau_{Aero y} + \tau_{Thrusty} + \tau_{Dif fy} \\ \tau_{Aero z} + \tau_{Thrustz} + \tau_{Diffz} \end{bmatrix} \quad (1.39)$$

1.8 Torques resulting from thrust

In hover or very-low-airspeed operation, the freestream aerodynamic forces scale toward zero with dynamic pressure, and become numerically ill-conditioned. Conventional fixed-wing stability-derivative models therefore cease to be a useful description of the aerodynamic forces. In the assumed craft geometry control surfaces are not immersed in rotor slipstream and thus provide near zero amount of control. In pure hover the craft relies instead on it's engines both to stay in the air and for control.

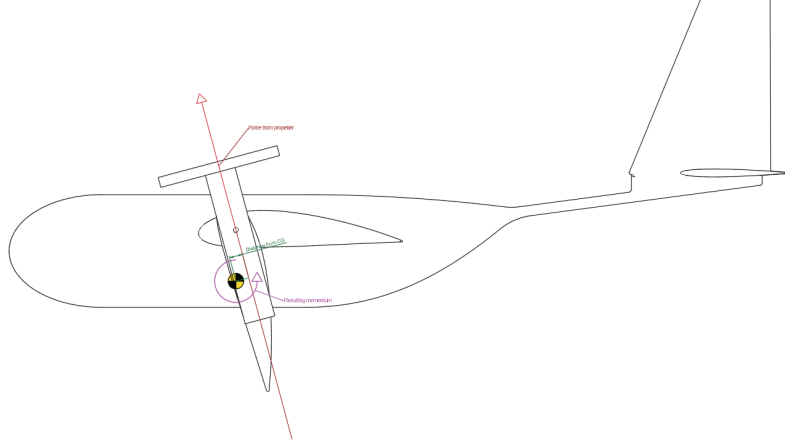


FIGURE 1.5: The side view of thrust creating a pitching torque.

The thrusts of engines are offset in z and y axes creating torques with their rotation. The component forces 1.9, 1.10 and 1.11 generated by the engines can be arranged into a vector and multiplied by vector of distances from CoM arranged into a skew-symmetric cross-product matrix:

$$\vec{\tau} = [\vec{r}] \times \vec{F} = \begin{bmatrix} 0 & -r_z & r_y \\ r_z & 0 & -r_x \\ -r_y & r_x & 0 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} \quad (1.40)$$

This can also be arranged into following form to derive component torques:

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix} \quad (1.41)$$

Due to the studied model being powered by two engines these must be expanded into:

$$\vec{\tau} = \vec{r}_r \times \vec{F}_r + \vec{r}_l \times \vec{F}_l = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{rx} & r_{ry} & r_{rz} \\ F_{rx} & F_{ry} & F_{rz} \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{lx} & r_{ly} & r_{lz} \\ F_{lx} & F_{ly} & F_{lz} \end{vmatrix} \quad (1.42)$$

The final form for the component forces is:

$$\vec{\tau} = \vec{r}_r \times \vec{F}_r + \vec{r}_l \times \vec{F}_l = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{rx} & r_{ry} & r_{rz} \\ F_{rx} & F_{ry} & F_{rz} \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{lx} & r_{ly} & r_{lz} \\ F_{lx} & F_{ly} & F_{lz} \end{vmatrix} \quad (1.43)$$

$$\tau_{Thrustx} = (E_{Span} * -(T(\delta N_r, \rho) \sin(\eta E_r))) + (E_{Span} * (T(\delta N_l, \rho) \sin(\eta E_l))) \quad (1.44)$$

$$\tau_{Thrusty} = (E_{Height} * (T(\delta N_r, \rho) \cos(\eta E_r))) + (E_{Height} * (T(\delta N_l, \rho) \cos(\eta E_l))) \quad (1.45)$$

$$\tau_{Thrustz} = -(E_{Span} * (T(\delta N_r, \rho) \cos(\eta E_r))) - (-E_{Span} * (T(\delta N_l, \rho) \cos(\eta E_l))) \quad (1.46)$$

For the assumed geometry in hover, the vertical offset between the thrust-vector rotation axis and the center of mass provides an important pitch-control moment when the thrust vector develops a longitudinal component. This offset results in rotation of the engine generating a torque as

showcased in 1.5. Without this offset the craft of the assumed configuration would be uncontrollable in pitch.

Besides of torque created by differential engine tilt the torque in the y axis is also affected by torque created by spinning propellers. This torque is described by equation:

$$Q = \rho * n^2 * D^5 * K_q \quad (1.47)$$

The model assumes counter-rotating propellers where the torque produced by both propellers will be opposed and will result in:

$$\tau_{Diffx} = (\rho * \delta N_r^2 * D_{prop}^5 * K_q) \cos \eta E_r - (\rho * \delta N_l^2 * D_{prop}^5 * K_q) \cos \eta E_l \quad (1.48)$$

$$\tau_{Difly} = 0 \quad (1.49)$$

$$\tau_{Diffz} = (\rho * \delta N_r^2 * D_{prop}^5 * K_q) \sin \eta E_r - (\rho * \delta N_l^2 * D_{prop}^5 * K_q) \sin \eta E_l \quad (1.50)$$

1.9 Torques resulting from aerodynamic forces

The torques acting on the craft in result of the aerodynamic forces in forms of control surface deflections and other aerodynamic effects can be described in standard stability analysis form:

$$\tau_{Aerox} = q_\infty S b (C_{l0} + C_{l\beta} \beta + C_{l\bar{p}} \bar{p} + C_{l\bar{r}} \bar{r} + C_{l\delta_A} \delta A + C_{l\delta_R} \delta R) \quad (1.51)$$

$$\tau_{Aero y} = q_\infty S \bar{c} (C_{m0} + C_{m\alpha} \alpha + C_{m\bar{q}} \bar{q} + C_{m\delta_E} \delta E) \quad (1.52)$$

$$\tau_{Aero z} = q_\infty S b (C_{n0} + C_{n\beta} \beta + C_{n\bar{p}} \bar{p} + C_{n\bar{r}} \bar{r} + C_{n\delta_A} \delta A + C_{n\delta_R} \delta R) \quad (1.53)$$

Where q_∞ is the dynamic pressure described as:

$$q_\infty = \frac{1}{2} \rho V_a^2 \quad (1.54)$$

And the non-dimensional angular rates are:

$$\bar{p} = \frac{pb}{2V_a} \quad (1.55)$$

$$\bar{q} = \frac{q\bar{c}}{2V_a} \quad (1.56)$$

$$\bar{r} = \frac{rb}{2V_a} \quad (1.57)$$

This form however results in singularities at $V_a = 0$ making the classical stability derivative models unusable in hover. The problematic V_a contained in the non-dimensional angular rates can be algebraically canceled with V_a appearing in the dynamic pressure equation resulting in following equations:

$$\tau_{Aerox} = \frac{1}{2} \rho V_a^2 S b (C_{l0} + C_{l\beta} \beta + C_{l\delta_R} \delta R + C_{l\delta_A} \delta A) + \frac{1}{4} \rho V_a S b^2 (C_{l_p} p + C_{l_r} r) \quad (1.58)$$

$$\tau_{Aero y} = \frac{1}{2} \rho V_a^2 S \bar{c} (C_{m0} + C_{m\alpha} \alpha + C_{m\delta_E} \delta E) + \frac{1}{4} \rho V_a S \bar{c}^2 (C_{m_q} q) \quad (1.59)$$

$$\tau_{Aero z} = \frac{1}{2} \rho V_a^2 S b (C_{n0} + C_{n\beta} \beta + C_{n\delta_A} \delta A + C_{n\delta_R} \delta R) + \frac{1}{4} \rho V_a S b^2 (C_{n_p} p + C_{n_r} r) \quad (1.60)$$

1.10 Angle and global position of the craft

To relate the velocities to crafts angles and global position requires mapping variables between a body-fixed local frame and an inertial global frame using rotation matrices and kinematic differential equations.

The global velocities used to differentiate the global positions will be obtained rotating the local velocity vector v_1 using the Direction Cosine Matrix denoted as R_{tb}

$$\dot{R}_1 = R_{tb}v_1 \quad (1.61)$$

Similarly a $J(\Theta)$ transformation matrix will be used to map the angular rates to Euler angle derivatives.

$$\dot{R}_2 = J^{-1}(\Theta)v_2 \quad (1.62)$$

This results in component equations for positions:

$$\dot{x}_E = u(\cos\theta\cos\psi) + v(\sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi) + w(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \quad (1.63)$$

$$\dot{y}_E = u(\cos\theta\sin\psi) + v(\sin\phi\sin\psi\sin\theta + \cos\phi\cos\psi) + w(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \quad (1.64)$$

$$\dot{z}_E = -u(\sin\theta) + v(\sin\phi\cos\theta) + w(\cos\phi\cos\theta) \quad (1.65)$$

And for rotations:

$$\dot{\phi} = p + q(\sin\phi\tan\theta) + r(\cos\phi\tan\theta) \quad (1.66)$$

$$\dot{\theta} = q(\cos\phi) - r(\sin\phi) \quad (1.67)$$

$$\dot{\psi} = q\left(\frac{\sin\phi}{\cos\theta}\right) + r\left(\frac{\cos\phi}{\cos\theta}\right) \quad (1.68)$$

1.11 Nonlinear state-space model

$$\begin{aligned}
\dot{\xi} = \begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \\ \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \\ \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \\ \eta \dot{E}_r \\ \eta \dot{E}_l \end{bmatrix} &= \begin{bmatrix} u(\cos\theta\cos\psi) + v(\sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi) + w(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \\ u(\cos\theta\sin\psi) + v(\sin\phi\sin\psi\sin\theta + \cos\phi\cos\psi) + w(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \\ -u(\sin\theta) + v(\sin\phi\cos\theta) + w(\cos\phi\cos\theta) \\ p + q(\sin\phi\tan\theta) + r(\cos\phi\tan\theta) \\ q(\cos\phi) - r(\sin\phi) \\ q(\frac{\sin\phi}{\cos\theta}) + r(\frac{\cos\phi}{\cos\theta}) \\ rv - qw + \frac{X}{m_{craft}} \\ pw - ru + \frac{Y}{m_{craft}} \\ qu - pv + \frac{Z}{m_{craft}} \\ c_1pq + c_2rq + c_7\tau_\phi + c_9\tau_\psi \\ c_3(p^2 - r^2) + c_4pr + c_8\tau_\theta \\ c_5pq + c_6rq + c_9\tau_\phi + c_{10}\tau_\psi \\ clip(\frac{1}{\tau_e}(\delta\eta E_r - \eta E_r), -\omega_{emax}, +\omega_{emax}) \\ clip(\frac{1}{\tau_e}(\delta\eta E_l - \eta E_l), -\omega_{emax}, +\omega_{emax}) \end{bmatrix} \quad (1.69)
\end{aligned}$$

Chapter 2

Select trims of the model

2.1 Hover trim

The trim at hover is the easiest case to analyze and provides clean closed form solutions. For symmetric, stationary hover take:

$$u = v = w = p = q = r = 0$$

$$\phi = \theta = 0$$

$$\eta E_r = \eta E_l = 90^\circ$$

In this condition $V_a = 0$ which causes the aerodynamic forces to vanish and aerodynamic moments tend to zero. This reduces finding trim at hover to finding engine command that counteracts the force of gravity.

With both engines at 90° the forces resulting from engine thrust become:

$$F_{Thrustx} = 0, F_{Thrustz} = -(T_r + T_l)$$

Since the body z -axis points downward, vertical equilibrium for a symmetric solution becomes:

$$T_r = T_l = \frac{mg}{2}$$

Therefor using thrust law as proposed at 1.8 the equilibrium becomes:

$$\delta\eta E_r = \delta\eta E_l = \sqrt{\frac{m_{craft}g}{2C_T\rho D_{prop}^4}}, \eta E_r = \eta E_l = \frac{\pi}{2} \quad (2.1)$$

2.2 Forward straight level flight trim

The symmetric straight-and-level trim problem assumes prescribed V_a and engine tilt. Prescribing these variables is required as otherwise the trim becomes undetermined.

The trim imposes following conditions:

$$p = q = r = 0$$

$$v = 0, \beta = 0, \phi = 0$$

$$\eta E_r = \eta E_l = \eta E$$

$$N_r = N_l = N$$

Let the prescribed airspeed be $V_a > 0$. From the definition of angle of attack:

$$u = V_a \cos \alpha$$

$$w = V_a \sin \alpha$$

Level flight requires $\dot{z}_E = 0$. The imposition $\phi = 0$ reduces the kinematics of $\dot{z}_E$ to:

$$\dot{z}_E = -u \sin \theta + w \cos \theta \quad (2.2)$$

Therefore:

$$-V_a \cos \alpha \sin \theta + V_a \sin \alpha \cos \theta = V_a \sin(\alpha - \theta) = 0 \quad (2.3)$$

For this trim thus $\theta = \alpha$. Once α is found the entire kinematic trim state is known:

$$u = V_a \cos \alpha, \quad v = 0, \quad w = V_a \sin \alpha, \quad \phi = 0, \quad \theta = \alpha, \quad p = q = r = 0 \quad (2.4)$$

2.2.1 Longitudinal force equilibrium and angle of attack

Keeping constant airspeed requires the thrust generated by the engines to be in equilibrium with aerodynamic forces opposing them. The symmetrical assumption of the trim allows for the total thrust to be assumed as:

$$P = T_r + T_l = 2T \quad (2.5)$$

The force of gravity can at $\theta = \alpha$ and $\phi = 0$ becomes:

$$mg = \begin{bmatrix} -\sin \alpha \\ 0 \\ \cos \alpha \end{bmatrix} \quad (2.6)$$

Since $p = q = r = 0$ keeping constant u and w requires the sums of forces X and Z to be both equal to 0 the two longitudinal force equations are:

$$P \cos \eta E - D \cos \alpha + L \sin \alpha - mg \sin \alpha = 0 \quad (2.7)$$

and

$$-P \sin \eta E - D \sin \alpha + L \cos \alpha + mg \cos \alpha = 0 \quad (2.8)$$

These could already be solved as a two-variable nonlinear system in (P, α) , but P can be further eliminated analytically by rotating the force balance into flight-path coordinates. Projecting 2.7-2.8 along the flight path results in:

$$P(\cos \eta E \cos \alpha - \sin \eta E \sin \alpha) - D = 0$$

thus:

$$P \cos(\eta E + \alpha) = D \quad (2.9)$$

Provided that $\cos(\eta E + \alpha) \neq 0$ this becomes:

$$P(\alpha) = \frac{D(\alpha)}{\cos(\eta E + \alpha)} \quad (2.10)$$

Projecting normal to the velocity results in the second equilibrium condition becoming:

$$mg - L - P \sin(\eta E + \alpha) = 0$$

thus:

$$L + P \sin(\eta E + \alpha) = mg \quad (2.11)$$

The P in this equation can be substituted with 2.10 resulting in:

$$L(\alpha) + D(\alpha)\tan(\eta E + \alpha) - mg = 0 \quad (2.12)$$

The aerodynamic forces here can be expanded resulting in final single scalar trim equation:

$$f(\alpha) = q_\infty S(C_l(\alpha) + C_d(\alpha)\tan(\eta E + \alpha)) - mg = 0 \quad (2.13)$$

This equation needs to be solved numerically to determine α and allow the full trim to be computed. An alternative form can also be used to avoid the explicit tangent:

$$\tilde{f}(\alpha) = (L(\alpha) - mg)\cos(\eta E + \alpha) + D(\alpha)\sin(\eta E + \alpha) = 0 \quad (2.14)$$

Once α^* is known P can be calculated as:

$$P^* = \frac{D^*}{\cos(\eta E + \alpha^*)} \quad (2.15)$$

where:

$$D^* = q_\infty S C_d(\alpha)(\alpha^*)$$

Therefor using thrust law as proposed at 1.8 the engine control input becomes:

$$\delta\eta E_r = \delta\eta E_l = \sqrt{\frac{P^*}{2C_T\rho D_{prop}^4}} \quad (2.16)$$

2.2.2 Pitch moment equilibrium

In order to satisfy the $q = 0$ condition for the proposed model the elevator has to cancel both aerodynamic pitch moment and moment the thrust induced pitch moment. The complete equation for pitch moment equilibrium is:

$$q_\infty S \bar{c}(C_{m_0} + C_{m_\alpha}\alpha^* + C_{m_{\delta_E}}\delta E) + E_{Height}P^*\cos\eta E = 0$$

thus:

$$\delta E = \frac{C_{m_0} + C_{m_\alpha}\alpha^* + \frac{E_{Height}P^*\cos\eta E}{q_\infty S \bar{c}}}{C_{m_{\delta_E}}} \quad (2.17)$$

Given that the $\dot{\eta E} = 0$ the tilt command becomes:

$$\delta\eta E_r = \delta\eta E_l = \eta E \quad (2.18)$$

2.2.3 Yaw and roll moment equilibrium

For an exactly symmetric state $\beta = p = r = 0$ and equal engine thrust/tilt makes both thrust-induced roll/yaw moments and the differential propeller moments cancel. For a case where the aerodynamic model is perfectly symmetric as well, that is C_{l_0} and C_{n_o} both equal to zero the rudder and aileron commands become:

$$\delta A = \delta R = 0$$

For other cases the trim can still be solved by solving:

$$C_{l_{\delta_A}}\delta A + C_{l_{\delta_R}}\delta R = -C_{l_0}$$

$$C_{n_{\delta_A}}\delta A + C_{n_{\delta_R}}\delta R = -C_{n_o}$$

Thus the control inputs become:

$$\delta A = \frac{-C_{l0}C_{n_{\delta_R}} + C_{l_{\delta_R}}C_{n_o}}{\Delta} \quad (2.19)$$

$$\delta R = \frac{-C_{l_{\delta_A}}C_{n_o} + C_{n_{\delta_A}}C_{l0}}{\Delta} \quad (2.20)$$

Where:

$$\Delta = C_{l_{\delta_A}}C_{n_{\delta_R}} - C_{l_{\delta_R}}C_{n_{\delta_A}}$$

2.3 Transition trim manifold